William Rohren

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Education

Texas A&M University - GPA: 3.15 *College Station, TX*

***BSME**: Bachelor of Science in Mechanical Engineering* **Graduation: Dec 2026**

Skills

Simulation & Analysis
Ansys FEA, Femap/Nastran, Topology Optimization, Python

Test & Instrumentation
Strain Gauges, Bridge Circuits, DAQ, Instron (MEEN 361)

Design & Fabrication
SolidWorks, CNC Machining, 3D Printing, Plasma Cutting

Electronics & Embedded Systems
Altium, KiCAD, Mixed-Signal PCB, SMT, Embedded C

Experience

Texas A&M SAE Aero Design Team *College Station, TX*

Product Owner, Structures (Officer) **Sep 2026 – Present**

- Co-lead a 9-engineer structures subteam, owning all structural design for the 2026-27 competition airframe
- Delegate tasks, hold the build schedule, and review subteam analysis for proper modeling assumptions

Structures & Materials Science (SMS) Engineer **May 2025 – Sep 2026**

- Owned the design of a cantilevered nose beam motor mount, consolidating propulsion electronics in the fuselage; designed 5 noses across 3 airframes, all flight tested with no structural failures under design loads
- Swept Ansys topology optimization from 5–50% mass retention, then reworked the 20% output into a buildable 9.1 oz balsa truss, FEA-verified under 25 lbf landing and 12 lbf per-motor thrust loads
- Scripted orthotropic balsa modeling in Ansys with Python, aligning each member's material axes to its grain; the nose's factor of safety dropped from 5.7 (isotropic) to 2.5, with cross-grain compression governing
- Modeled the main wing's 1/64 in leading-edge skin three ways (3D solids, 2D shells via midsurfacing, omitted) to set FEA fidelity; all agreed within 5%, so the team dropped skin from the model and cut mesh size and solve time

Rapid Prototyping Studio (RPS) *College Station, TX*

Student Technician **Jan 2024 – Present**

- Trained 500+ students on shop equipment and safety across FDM/SLA printing, laser cutting, and manual machining
- Maintained and repaired studio fabrication equipment, managing $14,000 in annual consumables and repairs
- Self-trained on CNC plasma cutting to bring an unused Torchmate table into service; authored the shop's 34-page SOP and trained 8 staff on the machine

Projects

Senior Capstone: Spacecraft Lifting Fixture (MGSE) - Southwest Research Institute **Aug 2025 – May 2026**

- Owned strain instrumentation and FEA correlation for a 6-person team designing a modular ISO 7 cleanroom lifting fixture rated for 200–1000 kg spacecraft
- Built a custom DAQ reading eight 350 Ω gauges in half-bridge pairs through four HX711 24-bit ADCs on an Arduino Nano, bonding gauges and fabricating harnesses on site
- Measured 76.2 and 71.8 $\mu\varepsilon$ against an Ansys prediction of 76.5 $\mu\varepsilon$ (0.4% and 6.2%) during a 1300 lb proof lift, using SolidWorks split lines to extract simulated strain at the exact gauge locations

Project SAVIOR - 16-Channel Strain Data Recorder **Apr 2026 – Present**

- Designing a 16-channel strain DAQ to measure in-flight wing loads against FEA predictions (eight ADS1220 24-bit ADCs, Teensy 4.1, 6-DoF IMU, ±200 g accelerometer); schematic drafted, layout, fab, and testing pending
- Selected bridge topology per channel by what each measurement must reject: half-bridge type II for bending, 3-wire series-pair arms for axial

Custom Atmega32U4 Development Board **Aug 2024 – Jan 2025**

- Designed a custom ATmega32U4 dev board in Altium, programmed bare-metal in C, manufactured via JLCPCB
- Debugged USB-C implementation failures across two board revisions; successfully flashed firmware via ICSP

Additional Interests

Astro-Photography, Fishing, Trumpet, PC-Building, 3D Printing, Embedded Systems, RC Aircraft